FIN3080: Investment Analysis and Portfolio Management

SCHOOL OF MANAGEMENT AND ECONOMICS

Instructor Prof. Jinfan Zhang

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Introduction



教育背景：

- 耶鲁大学金融经济学博士，2013
- 哈佛大学统计学硕士，2008
- 清华大学电子工程学博士，2007
- 清华大学电子工程学硕士，2005
- 清华大学电子工程学本科，2003
- 北京大学经济学本科双学位，2006

工作背景：

- 教育部青年长江学者
- 香港中文大学深圳经济管理学院终身教授
- 香港中文大学深圳经济管理学院金融系主任
- 中国证监会中证研究院特聘研究员（2022年3月-至今）
- 国际货币基金组织IMF经济学家(2014-2017)
- 中国人民银行研究局高级访问学者（2019年6月-9月）

Class information

- E-mail zhangjinfan@cuhk.edu.cn
- Grading
  - Attendance 10%
  - Four projects 40% Coding! Coding! Coding! (UCLA: https://stats.oarc.ucla.edu/stata/modules/)
  - Final exam 50%
- Reading
  - Equity Asset Valuation (股权资产估值) Jerald Pinto etc.
  - Bodie, Zvi, Alex Kane, and Alan J. Marcus ("BKM"). *Investments*.
  - 投资中最简单的事（邱国鹭）
  - 危机的逻辑（王松，张劲帆）

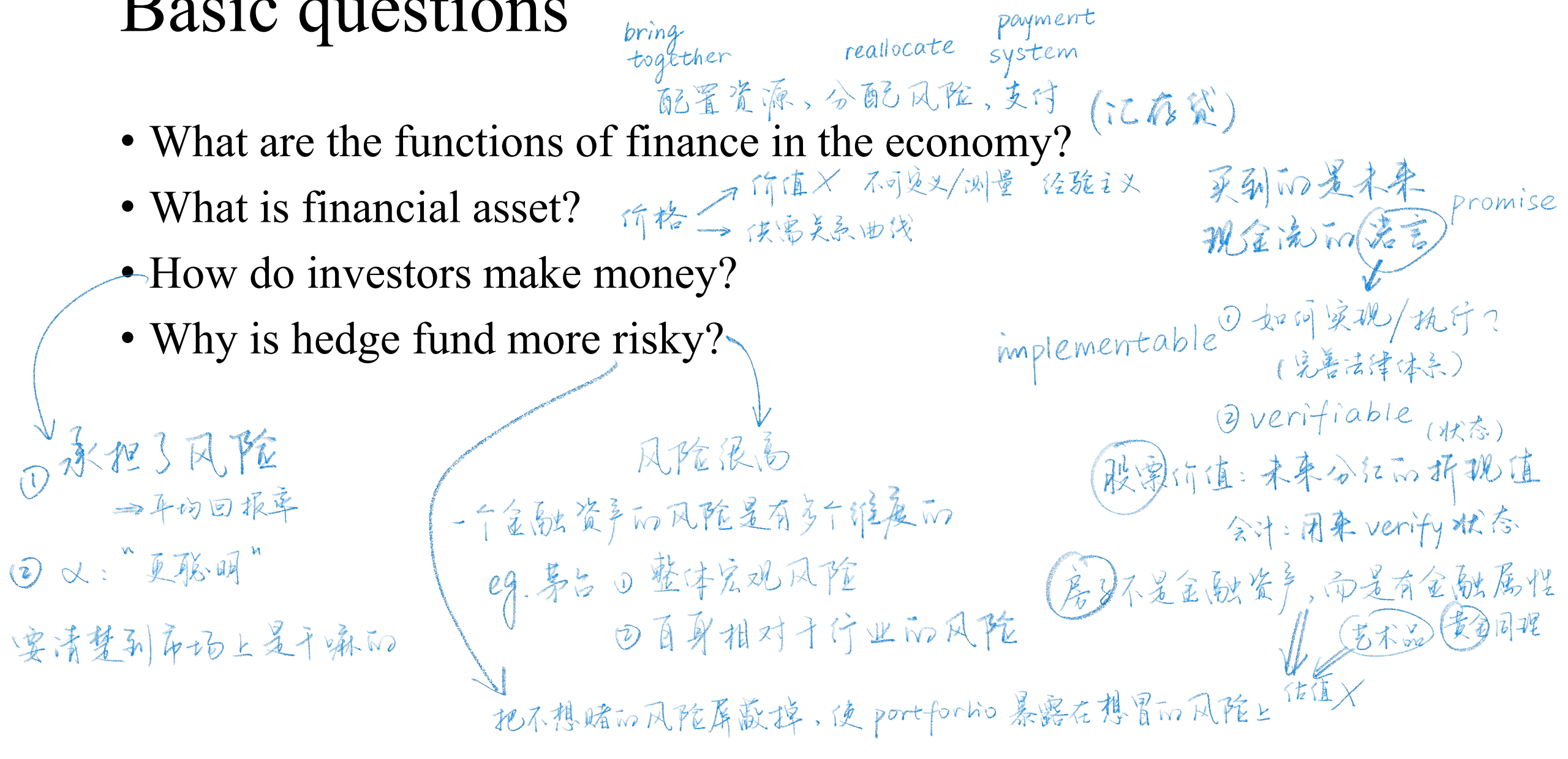
林彪元帅说学习毛主席语录要“带着问题学，活学活用，学用结合，急用先学，立竿见影”。

Class information

- TA
  - Jingyu Bi(毕敬禹) jiayizheng@cuhk.edu.cn, Zhiren Building Room 320 or Zoom
  - Kaixuan Zheng (郑凯轩) kaixuanzheng@cuhk.edu.cn, Zhiren Building Room 319 or Zoom
  - Sijie Wang (王思杰) 218012036@link.cuhk.edu.cn,
  - Three student TAs, who all received A in last year.

Basic questions

- What are the functions of finance in the economy?
- What is financial asset?
- How do investors make money?
- Why is hedge fund more risky?



Basic questions

- What are the functions of finance
  - Investment & Financing (the two sides of the same coin)
  - Risk reallocation (futures, insurance)
  - Payment
- What is financial asset?
  - Financial assets vs real goods
    - A promise of future state dependent cash flow
  - Verifiable and implementable
    - Accounting
    - Rule of law and other mechanism
- How do investors make money?
  - Beta vs alpha
  - Efficient market or not?
- Why is hedge fund more risky?
  - Hedge means reducing risk, but why hedge fund is more risky?
  - Hedge more bet more

Roadmap

- Valuation of individual stock
  - Intrinsic valuation
  - Relative valuation
- Portfolio analysis
  - Risk return analysis
- Behavioral finance
  - Limit to arbitrage
  - Psychology
- Institutional concern
  - Corporate Governance

Approaches to Valuation

- **Intrinsic valuation**, relates the value of an asset to its intrinsic characteristics: its capacity to generate cash flows and the risk of the cash flows. In its most common form, intrinsic value is computed with a discounted cash flow valuation, with the value of an asset being the present value of expected future cash flows on that asset.

- **Relative valuation or pricing**, estimates the value of an asset by looking at the pricing of 'comparable' assets relative to a common variable like earnings, cashflows, book value or sales.

Misconceptions about Valuation

- Myth 1: A valuation is an objective search for true value
  - Truth 1.1: All valuations are biased. The only questions are "how much" and in which direction.
- Myth 2: A good valuation provides a precise estimate of value
  - Truth 2.1: There are no precise valuations.
  - Truth 2.2: The payoff to valuation is greatest when valuation is least precise.
- Myth 3: The more quantitative a model, the better the valuation
  - Truth 3.1: One's understanding of a valuation model is inversely proportional to the number of inputs required for the model.
  - Truth 3.2: Simpler valuation models do much better than complex ones.

Basis for all valuation approaches

- The use of valuation models in investment decisions (i.e., in decisions on which assets are under-valued and which are over-valued) are based upon
  - a perception that markets are inefficient and make mistakes in assessing value
- In an efficient market, the market price is the best estimate of value. The purpose of any valuation model is then the justification of this value.

Discounted Cash flow Valuation (DCF)

- What is it: In discounted cash flow valuation, the value of an asset is the present value of the expected cash flows on the asset.
- Philosophical Basis: Every asset has an intrinsic value that can be estimated, based upon its characteristics in terms of cash flows, growth and risk.
- Information Needed: To use discounted cash flow valuation, you need
  - to estimate the life of the asset
  - to estimate the cash flows during the life of the asset
  - to estimate the discount rate to apply to these cash flows to get present value
- Market Inefficiency: Markets are assumed to make mistakes in pricing assets across time, and are assumed to correct themselves over time, as new information comes out about assets.

Advantages of DCF Valuation

- Since DCF valuation, done right, is based upon an asset's fundamentals, it should be less exposed to market moods and perceptions.
- If good investors buy businesses, rather than stocks (the Warren Buffet adage), discounted cash flow valuation is the right way to think about what you are getting when you buy an asset.
- DCF valuation forces you to think about the underlying characteristics of the firm, and understand its business. If nothing else, it brings you face to face with the assumptions you are making when you pay a given price for an asset.
- If you buy into the notion of value being driven by a company's cash flows, you are immunized (to the extent that you have a long time horizon) from what the market thinks about your investment..

Disadvantages of DCF valuation

- Since it is an attempt to estimate intrinsic value, it requires far more explicit inputs and information than other valuation approaches
- These inputs and information are not only noisy (and difficult to estimate), but can be manipulated by the analyst to provide the conclusion he or she wants. The quality of the analyst then becomes a function of how well he or she can hide the manipulation.
- In an intrinsic valuation model, there is no guarantee that anything will emerge as under or over valued. Thus, it is possible in a DCF valuation model, to find every stock in a market to be over valued. This can be a problem for
  - equity research analysts, whose job it is to follow sectors and make recommendations on the most under and over valued stocks in that sector
  - equity portfolio managers, who have to be fully (or close to fully) invested in equities

When DCF Valuation works best

- At the risk of stating the obvious, this approach is designed for use for assets (firms) that derive their value from their capacity to generate cash flows in the future.
- It works best for investors who
  - have a long time horizon, allowing the market time to correct its valuation mistakes and for price to revert to "true" value or
  - are capable of providing the catalyst needed to move price to value, as would be the case if you were an activist investor or a potential acquirer of the whole firm
  - are not easily swayed or affected by market movements that are contrary to their "value" views

Relative Valuation (Pricing)

- What is it?: The value of any asset can be estimated by looking at how the market prices "similar" or 'comparable" assets.
- Philosophical Basis: The intrinsic value of an asset is impossible (or close to impossible) to estimate. The price of an asset is whatever the market is willing to pay for it (based upon its characteristics)
- Information Needed: To do a relative valuation, you need
  - an identical asset, or a group of comparable or similar assets
  - a standardized measure of value (in equity, this is obtained by dividing the price by a common variable, such as earnings or book value)
  - and if the assets are not perfectly comparable, variables to control for the differences
- Market Inefficiency: Pricing errors made across similar or comparable assets are easier to spot, easier to exploit and are much more quickly corrected.

Advantages of Relative Valuation

- In sync with the market: Relative valuation is much more likely to reflect market perceptions and moods than discounted cash flow valuation. This can be an advantage when it is important that the price reflect these perceptions as is the case when
  - the objective is to sell an asset at that price today (IPO, M&A)
  - investing on "momentum" based strategies
- With relative valuation, there will always be a significant proportion of securities that are under valued and over valued. Since portfolio managers are judged based upon how they perform on a relative basis (to the market and other money managers), relative valuation is more tailored to their needs
- Relative valuation generally requires less explicit information than discounted cash flow valuation.

Disadvantages of Relative Valuation

- A portfolio that is composed of stocks which are under valued on a relative basis may still be overvalued, even if the analysts' judgments are right. It is just less overvalued than other securities in the market.
- Relative valuation is built on the assumption that markets are correct in the aggregate, but make mistakes on individual securities. To the degree that markets can be over or under valued in the aggregate, relative valuation will fail
- Relative valuation may require less information in the way in which most analysts and portfolio managers use it. However, this is because implicit assumptions are made about other variables (that would have been required in a discounted cash flow valuation). To the extent that these implicit assumptions are wrong the relative valuation will also be wrong.

When relative valuation works best

- This approach is easiest to use when
  - there are a large number of assets comparable to the one being valued
  - these assets are priced in a market
  - there exists some common variable that can be used to standardize the price
- This approach tends to work best for investors
  - who have relatively short time horizons
  - are judged based upon a relative benchmark (the market, other portfolio managers following the same investment style etc.)
  - can take actions that can take advantage of the relative mispricing; for instance, a hedge fund can buy the under valued and sell the over valued assets

Portfolio analysis

- The benefit of diversification
- The risk and return balance
  - Sharpe ratio
- To construct the optimal portfolio in certain sense
  - The green portfolio
- To develop the quantitative strategies
  - Momentum
  - Value
  - Size

Behavioral Finance

- The fundamental debate in finance
  - Market is efficient or inefficient
- Limit to arbitrage
  - Short sale constraint
  - Short investment horizon
- Psychology
  - Overconfident
  - Herding
  - Mental accounting